Runbook: Setting up Databricks on AWS

Prerequisites

- An AWS account with sufficient permissions to create resources
- Databricks account with an active subscription
- A compatible AWS region (check the Databricks documentation for supported regions)
- A VPC and subnet in the desired AWS region

Step-by-Step Instructions

Step 1: Create an AWS EC2 Key Pair

- Log in to the AWS Management Console and navigate to the EC2 dashboard
- Click on "Key pairs" in the left-hand menu and then "Create key pair"
- Choose a name for the key pair and select "RSA" as the key type
- Click "Create key pair" and download the .pem file

Step 2: Create an AWS VPC and Subnet

- Log in to the AWS Management Console and navigate to the VPC dashboard
- Click on "Subnets" in the left-hand menu and then "Create subnet"
- Choose the desired VPC and CIDR block for the subnet
- Set the availability zone and click "Create subnet"

Step 3: Create an AWS Security Group

- Log in to the AWS Management Console and navigate to the VPC dashboard
- Click on "Security groups" in the left-hand menu and then "Create security group"
- Choose the VPC and subnet created in Step 2
- Add rules for inbound traffic (e.g., 22 for SSH) and outbound traffic (e.g., 8443 for Databricks)
- Click "Create security group"

Step 4: Launch an AWS EC2 Instance

- Log in to the AWS Management Console and navigate to the EC2 dashboard
- Click on "Instances" in the left-hand menu and then "Launch instance"
- Choose the desired instance type and operating system (e.g., Ubuntu)
- Configure the instance details, including the VPC, subnet, and security group created in previous steps
- Launch the instance and wait for it to become available

Step 5: Configure the Databricks Cluster

- Log in to the Databricks dashboard and navigate to the "Clusters" page
- Click on "Create cluster" and choose the "AWS" platform
- Enter the desired cluster name, instance type, and node count
- Configure the cluster settings, including the VPC, subnet, and security group created in previous steps
- Click "Create cluster" to launch the cluster

Step 6: Install and Configure the Databricks Driver

- Log in to the EC2 instance and install the Databricks driver using the following command: sudo curl -sS https://databricks.com/files/databricks-driver-aws-latest.zip -o /tmp/databricks-driver.zip && sudo unzip /tmp/databricks-driver.zip -d /usr/local/databricks-driver
- Configure the Databricks driver by creating a file /etc/databricks-driver.conf with the following contents: [driver] aws_access_key_id = <your_access_key> aws_secret_access_key = <your_secret_key> aws_region = <your_region>

Step 7: Start the Databricks Cluster

- Log in to the Databricks dashboard and navigate to the "Clusters" page
- Click on the cluster name to view its details and click the "Start" button to start the cluster

Expected Outcomes

- A Databricks cluster is created and running on an AWS EC2 instance
- The cluster is configured to use the VPC, subnet, and security group created in previous steps
- The Databricks driver is installed and configured on the EC2 instance

Troubleshooting Tips

- If the cluster fails to start, check the instance logs for errors and ensure that the instance is running and has sufficient resources
- If the Databricks driver installation fails, check the installation logs for errors and ensure that the instance has sufficient permissions to install the driver
- If you encounter issues with network connectivity, check the security group rules and ensure that the necessary ports are open

Links to Relevant Documentation

- Databricks documentation: https://docs.databricks.com/
- AWS documentation: https://docs.aws.amazon.com/

Note: This runbook assumes a basic understanding of AWS and Databricks concepts. If you are new to these technologies, it is recommended that you complete the relevant tutorials and documentation before attempting to set up a Databricks cluster on AWS.